Chapter No 8 – Decision

C E – 101 : C & P F

Computing and Programming Fundamentals

Batch 2014

Compiled By:

Syed Atir Iftikhar

Sir Syed University of Engineering & Technology
Computer Engineering Department
University Road, Karachi-75300, PAKISTAN

CE - 101 : Computing and Programming Fundamentals

- **Course Objectives:**
  - Upon successful completion of this course, the student will be able to:
    - Fundamentals of Computer Engineering and Information Technology
      - Overview of History, Classification and Components of Computers
      - Number Systems and Logic Gates
      - Overview of Software, Operating Systems, Networks and Internet
    - Introduction about Programming
      - Basic Building Blocks, Loop, Decision Making

2 0 1 4

CPF Books

- **Text Book:**

  1. Introduction to Computers (7th Edition)
  By Peter Norton

  2. Turbo C Programming For The PC (Revised Edition)
  By Robert Lafore

- **Reference Books:**

  1. Computer, Communications and Information.
  By Sarah Hutchinson and Stacey Sawyer

  2. Let Us C
  By Yashavant Kanetkar

2014

Marks Distribution

Mid Term	20
Lab Work + Project	20
Quiz	5
Assignment + Class Performance	5
Semester Final Paper	50
Total Marks	**100**

- ***Performance Bonus*** ______________ ***5 Marks***

2014

Course Instructors

- **Syed Aatir Iftikhar** satirs@hotmail.com

  Lecturer, CED

  Room No: BF-03

  Section B

2 0 1 4

7

Course Outline

✓ PART A

- **History of Computer**
- **Classification of Computer**
- **Basic Components**
- **CPU, I/O, Peripheral Devices, Storage**
- **Von Neumann Architecture**
- **Number Systems**
- **Binary Numbers**
- **Boolean Logic**
- **Software**
- **Operating Systems**
- **Networks and Internet**

✓ PART B

- **Types of Programming Languages**
- **Algorithm**
- **Flow Chart**
- **Introduction to C Language Programming**
- **Basic Building Blocks**
- **Loops**
- **Decision Making Statements**

2 0 1 4

8

CE – 101: Computing and Programming Fundamentals

Chapter 8

Batch 2014

Decision

Compiled By: Syed Atir Iftikhar satirs@hotmail.com

Chapter 8 : Contents

- If statement
- If–else statement
- Switch statement
- Conditional Operator

2014

Introduction

- Computer Languages too must be able to perform different sets of actions depending on the circumstances.
- C Language has three major decision making structures: i.e.
  1. If Statement
  2. If-Else Statement
  3. Switch Statement

2014

If Statement

- Like most Languages, C Language uses the keyword **if** to introduce the basic decision making statement.

2014

Program Using If Statement

```
void main (void)
 {
    char ch;
    ch = getche( );
    if ( ch  = =  ‘ y ‘  )
      printf ( “ \n You typed y. “ );
 }
```

2014

Program Using If Statement

Output:

y

You typed y.

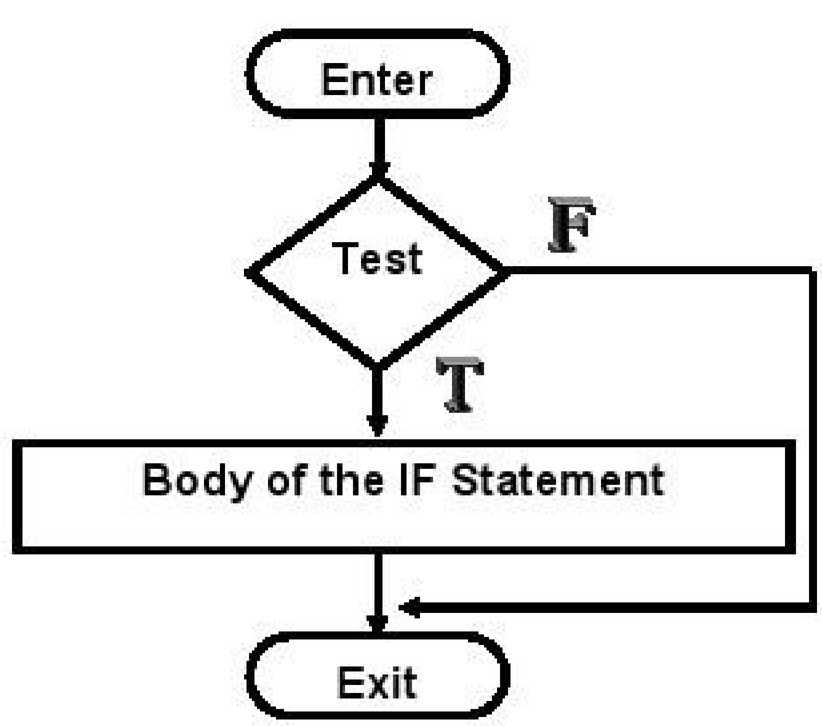
- If you type ‘ y ‘, the program will print “ You typed y “. If you type some other character, such as ‘ n ‘ the program does not do anything.

2014

Structure of IF Statement

- The **if** keyword is followed by the parenthesis, which contain a conditional expression using a relational operator.
- Following this, there is the body of the statement, consisting of either a single statement terminated by a semicolon or multiple statements enclosed by braces.
- The if statement is similar to the while statement in operation as well as format. In both cases the statements making up the body of the statement will not be executed at all, if the condition is false.

Operation of the If Statement



Word Counting Program

```
void main (void)
  {
   int charcnt  = 0 ;
   int wordcnt = 0 ;
   char ch ;
   clrscr( );
   printf ( " Type in a phrase :   " );
```

2014

Word Counting Program

```
while ( ( ch = getche ( )  ) ! = ' \r ' )
   {
     charcnt ++ ;

     if  ( ch = =  '   '  )
        wordcnt ++ ;
   }
   printf ( " \n " );
   printf ( " \n Character count is %d ", charcnt ) ;
   printf ( " \n Word count is %d ",  wordcnt + 1 ) ;
 }
```

2014

Word Counting Program

- **Output:**

  Type in a phrase : I Love Pakistan
  Character count is 15
  Word count is 3

- The statement wordcnt ++ ; causes the variable wordcnt to be incremented every time a space is detected in the input stream.
- There is always be one word than there are spaces between them, so we add 1 to the variable wordcnt before printing it out.

2014

Multiple Statements in IF Statement

- The body of the **IF** statement may consist of either a single statement terminated by a semi colon or by a number of statements enclosed in the braces. i.e delimiters.

2014

Program Using Multiple IF Statement

```
void main (void)
  {
        char ch;
        ch = getche( );
        if  ( ch = = ‘ y ‘ )
         {
                printf ( “ \n You typed y. “ );
                printf ( “ \n Not some other letter. “ );
         }
  }
```

2
0
1
4

Program Using Multiple IF Statement

- **Output:**

  y

  You typed y.

  Not some other letter.

2
0
1
4

Nested IF Statements

- Nested IF statements means that one if statement is a part of the body of another if statement.

2014

Program Using Nested IF Statements

```
void main (void)
  {
       clrscr( );
       if ( getche ( )  = =  ' n ' )
       if ( getche ( )  = =  ' o ' )
          printf ( " \n You typed no. " )
  }
```

2014

Program Using Nested IF Statements

- **Output:**

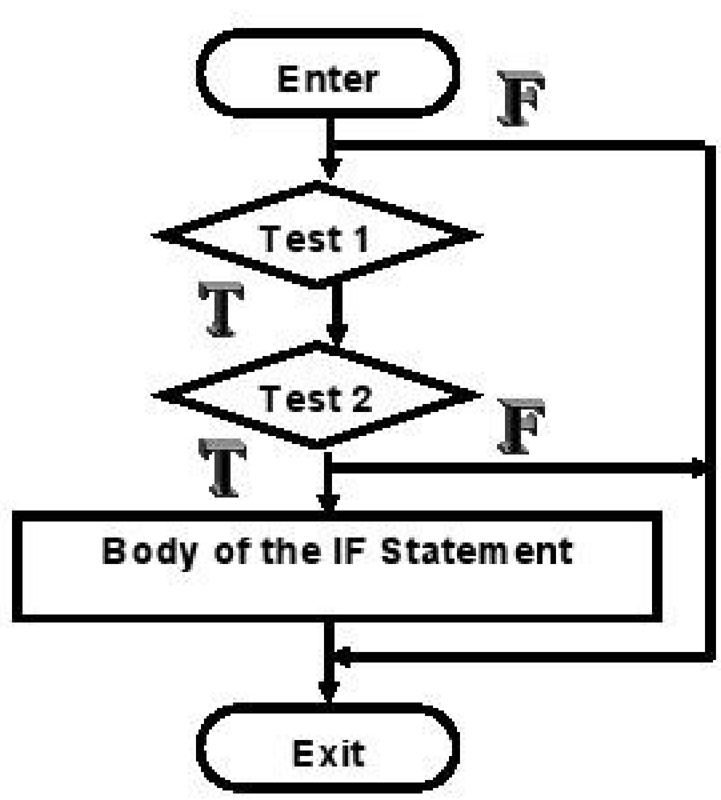
  no

  You typed no.

- In the above program, the inner if statement will not be reached unless the outer one is true, and the printf ( ) statement will not be executed unless both if statements are true.

2 0 1 4

Operation of the Nested - If Statement



2 0 1 4

The IF – Else Statement

- The if statement by itself will execute a single statement, or a group of statements, when the test expression is true. It does nothing when it is false.
- Can we execute a group of statements if and only if the test expression is not true?
- Of Course, this is the purpose of the if-else statement.

2014

Program using IF – Else Statement

```
void main (void)
  {
        char ch;
        ch = getche( );

        if  (  ch   = =   ‘ y ‘  )
        printf ( “ \n You typed y. “ );
        else
        printf ( “ \n You did not typed y. “ );
  }
```

2014

Program using IF – Else Statement

- **Output:**

a

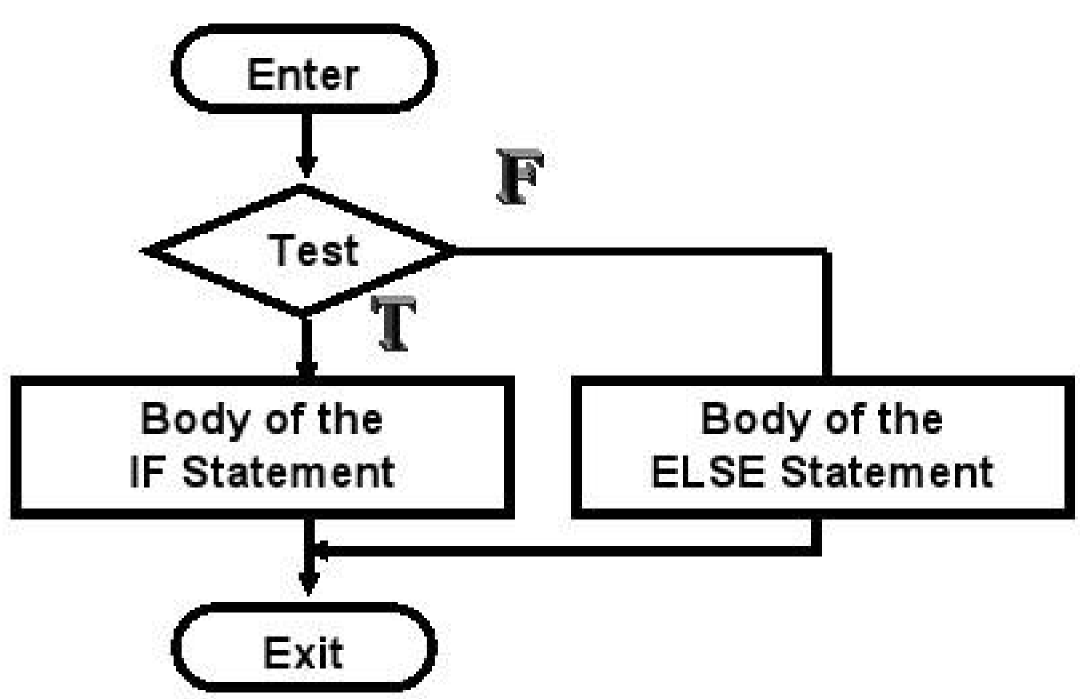
You did not typed y.

y

You typed y.

2 0 1 4

Operation of the If - Else Statement



2 0 1 4

Which IF gets the else ?

```
void main (void)
  {
   int temp ;  clrscr ( ) ;
   printf ( " Type today's temperature. " ) ;
   scanf ( " %d ", &temp ) ;
   if ( temp < 80 )
     if ( temp > 60 )
       printf ( " Nice Day." ) ;
   else
       printf ( " Sure is Hot! " ) ;
  }
```

2014

31

CPF - Chapter No 8 : Decision

Which IF gets the else ?

- **Output:**

  Type today's temperature. 32

- Suppose temp is 32. What will be printed when its program is executed ?
- Would you guess nothing at all ?
- That's seems to be reasonable: because the first if condition (temp < 80) will be true, so the second if condition (temp > 60) will be evaluated.
- It will be false so it looks as if the program will exit from the entire nested if-else construct.

2014

32

CPF - Chapter No 8 : Decision

Which IF gets the else ?

```
void main (void)
  {
   int temp ;
   printf ( " Type today's temperature. " ) ;
   scanf ( " %d ", &temp ) ;
   if ( temp < 80 )
      if ( temp > 60 )
        printf ( " Nice Day." ) ;
     else
        printf ( " Sure is Chilly! " ) ;
  }
```

2014

Which IF gets the else ?

- **Output:**

  Type today's temperature. 32

- Here the inner else is paired with the inner if, and the outer else is paired with the outer if. The indentation in this case is not misleading.
- **" An else is associated with the last if that does not have its own else."**
- Surrounding the if with braces makes it invisible to the else, which then matches up with the earliest non-braced if statement.

2014

Which IF gets the else ?

For Example:

```
if ( temp < 80 )
   {
      if ( temp > 60 )
         printf ( " Nice Day." ) ;
   }
else
         printf ( " Sure is Hot! " )
```

2014

35

CPF - Chapter No 8 : Decision

What is the Output?

```
int x = 5 , y = 3;


if ( x = 6 )
    printf (" x is 5 ");
if ( 0 <=  x  <= 4 )
    printf (" %d\n ", x);
else ;
    printf (" %d \n ", y);
```

2014

36

CPF - Chapter No 8 : Decision

Nested if-else-if

- This series of is commonly written as statements:

```
if( expression1 )
    statement1;
else
    if( expression2 )
        statement2;
    else
        if( expression3 )
            statement3;
        else
            statement4;
```

```
if( expression1 )
    statement1;
else if( expression2 )
    statement2;
else if( expression3 )
    statement3;
else
    statement4;
```

2014

Logical Operators

- Logical Operators are powerful way to condense and clarify complicated if-else structured.
- There are **3 types of Logical Operators** or sometimes called the **Boolean Operators** in C Language.

1.	||	**Logical OR**
2.	&&	**Logical AND**
3.	!	**Logical NOT**

2014

Program

```
void main (void)
  {
   int x ;  clrscr ( ) ;
   printf ( " Press the number 5 or 6 :  " ) ;
   scanf ( " %d ", &x ) ;
   if ( x = = 5 | | x = = 6 )
     printf ( "You have typed the right number. " ) ;
   else
     printf ( " Sorry wrong number. " ) ;
  }
```

2014

39

Program

- **Output:**

  2

  Sorry wrong number.

  5

  Yes You have typed the right number.

- The logical operator OR i.e (||) means if either the expression on the right side of the operator, or the expression on the left is true, then the entire expression is valid or true.

2014

40

Program

```
void main (void)
  {
   char x , y ;  clrscr( );
   scanf ( " %c ", &x );
   scanf ( " %c ", &y );
   if ( x = =  ' n '   &&   y = = ' o ' )
   printf ( " You typed no. " )
  }
```

Output:
no
You typed no.

2014

41

Program

```
void main (void)
  {
   int charcnt, digitcnt ;  clrscr( );
   printf ( " Type in a phrase :    " ) ;
   while (  ( ch = getche ( )  )  ! =  ' \r ' " ) ;
     {
        charcnt + + ;
        if ( ch > 47 && ch < 58 )
          digitcnt + + ;
     }
   printf ( " \n Character count is %d " , charcnt ) ;
   printf ( " \n Digit count is %d " , digitcnt ) ;
  }
```

2014

42

Program

- **Output:**

  Type in a phrase : Ali has 1 bike and 2 cars
  Character count is 25
  Digit count is 2

- The key to this program is the **logical AND operator.**
- This operator says:
  - if both the expression on the left (ch > 47) and (ch < 58) are true, then the entire expression (ch > 47 && ch 58) is true. This will only be true if ch is between 48 and 57; these are the ASCII codes for the digits from 0 to 9.

Precedence Check

- **" Logical Operator have a *lower* precedence than the Relational Operator. "**
- The Logical Operator (!) i.e NOT, is represented the exclamation print (!) and sometimes called the " Bang Operator ".
- This operator reverse the logical values of the expression it operates on, it makes a true expression false and a false expression true.
- The NOT Operator is a unary operator on, that is it takes only one operand.
- Example of the NOT operator applied to a relational expression: ! (x < 5)
- This means that " not x less than 5 ". In other words, if x is less than 5, the expression will be false, since (x < 5) is true.

The Switch Statement

- The Switch statement provides an alternative to the else – if statement construct.
- It is similar to the else – if statement, but has more flexibility and a clearer format.

2014

Structure of the Switch Statement

- Structurally, the statement starts out with the keyword **switch**, followed by the parenthesis containing an integer or character variable which we will call the **switch variable**.

2014

Structure of the Switch Statement

```
switch ( op )
  {
   case ' a ' :
            statement ;
            break ;

   case ' b ' :
            statement ;
            break ;
   default :
            statement ;
  }
```

2 0 1 4

Break and Default

- **Break** command is often useful when a condition suddenly occurs that makes it necessary to leave a loop before the loop expression becomes false.
- **Default** command is used if the switch variable does not match any of the case constants.

2 0 1 4

Program using Switch Statement

```
void main (void)
 {
  float num1 = 1.0 , num2 = 1.0 ;
  ch op ;     clrscr ( ) ;
  while ( ! ( num1 = = 0.0   &&   num2 = = 0.0 ) )
   {
    printf ( " Type number ,operator , number :  \n " ) ;
    scanf ( " %f  %c  %f ", &num1, &op, &num2 ) ;
```

2014

49

Program using Switch Statement

```
switch ( op )
 {
  case ' + ' :
            printf ( " = %f ", num1 + num2 ) ;
                   break ;
  case ' - ' :
                   printf ( " = %f ", num1 - num2 ) ;
            break ;
  case ' * ' :
            printf ( " = %f ", num1 * num2 ) ;
            break ;
```

2014

50

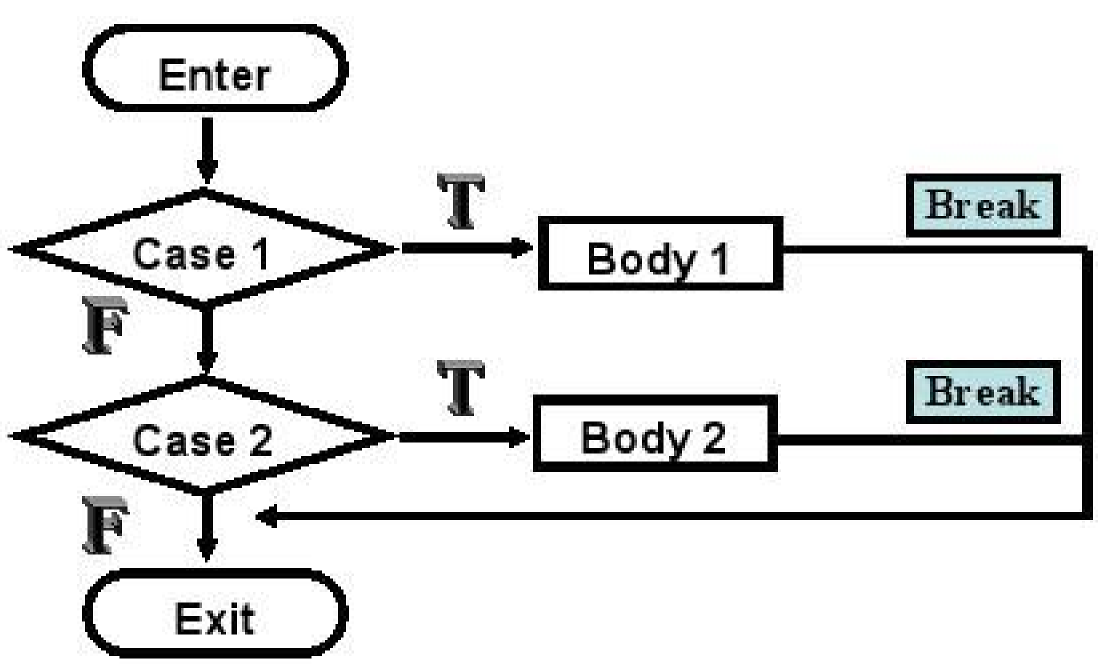
Program using Switch Statement

```
    case ' / ' :
                printf ( " = %f ", num1 / num2 ) ;
                break ;
    default :
                printf ( " Unknown Operator " ) ;
      }
printf ( " \n \n " );
      }

   getch ( );
}
```

2014

Operation of the Switch Statement



2014

What is the Output?

```
/* print the color*/

color  =  'R';
switch (color)
 {
        case 'R':
                printf ("Red \n");
        case 'B':
                printf ("Blue \n");
        case 'Y':
                printf ("Yellow \n");
 }
```


The Conditional Operator

- In C Language conditional operator is also called the ***Decision-making operator***.
- It consists of two symbols used on three different expressions, and thus it has the distinction of being the only ternary operator in C. (Ternary operators works on three variables, as opposed to the more common binary operators, such as (+), which operates on two expressions.

The Conditional Operator

- The conditional operator has the form:

  Condition ? expression1 : expression2.

- The conditional operator consists of both the question mark and the colon.
- Condition is a logical expression that evaluates to either true or false, while expression1 and expression2 are either values or expression that evaluates to values.

2014

The Conditional Operator

- The condition is evaluated. If it is true, then the entire conditional expression takes on the value of expression1. If it is false, the conditional expression takes on the value of expression2.
- Note that the entire conditional expression – the three expressions and two operators takes on a value and can therefore be used in a assignment statement.
- Here is an example:

  max = (num1 > num2) ? num1 : num2 ;

2014

The Conditional Operator

- This statement is equivalent to the if-else statement:

```
if ( num1 > num2 )
    max = num1;
else
    max = num2;
```

- In short, Conditional Operator returns one or the other of the two values, depending on whether a condition is true or false.

2014

Class Assignment No 4

1. Differentiate between if statement and if-else statement with respect to some suitable example.
2. Under which circumstances, switch statement is feasible.
3. Draw a flow chart of the program in which two numbers has been taken from the user.
   The program can perform the arithmetic operations like addition, subtraction, multiplication and division depending upon the user choice.

2014